UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement Native 3.3V MOSFET SPICE Model Document

Version 0.3 Phase 1
2013/03/04

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1 Introduction

1.1 Revision history

Version	Date	Owner	Comment
V0.1P1	2010/01/08	Jian Zhi Huang	1. Copy model from L110E_NVT_33_V011.lib in the G-05-LOGIC/MIXED_MODE11-MMC/L110E-SPICE 2. DSM version definition from 2009/11/30.
V0.2P1	2010/07/26	Ruei Chih Yen	Base on new silicon data.
V0.3P1	2013/03/04	Yealing Lin	Revise corner range.

1.2 General information

This document serves as a reference to the design works of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following device is presented in this document.

3.3V n-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.24

For detailed descriptions of this compact model equations and parameters, please refer to this material.

BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24, there are two types of models that could be used. One is binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is global model, which means one scalable parameter set is provided and it could cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for its many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the hardware data, which can be found in the following sections.

1.3 Model usage

The compact model is provided in various formats. It's guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2012.06
Cadence Spectre version 10.1.1.111.isr8
Mentor Graphics Eldo version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE
- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9

- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib" CaseName
, where CaseName could be TT, FF, or SS

For Cadence Spectre
- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9

- Add the following statement in the input file to the simulator.
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName could be TT, FF, or SS

For Mentor Graphics Eldo
- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9

- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib.eldo53" CaseName
, where CaseName could be TT, FF, or SS
For 3.3V n-ch MOSFET: n_nvt_33_l110e

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range:(After 90% shrink)
For 3.3V n-ch MOSFET:
1.632 µm <= L_{DES} <= 50 µm
0.9 µm <= W_{DES} <= 100 µm

Voltage range:
For 3.3V n-ch MOSFET:
0V <= V_{GS} <= 3.3V (*1.1)
0V <= V_{DS} <= 3.3V (*1.1)
-3.3V <= V_{BS} <= 0V

Temperature range:
For both 3.3V n-ch MOSFET:
-50 ℃ ~ +125 ℃

UMC L110E is UMC L130E 90% direct shrunk option.The following table give users (including designers) a dimension correlation between Drawing vs. SPICE/Resistor dimensions.

Feature\ GDSII	Drawing in L130E GDSII database (unit : um)	L110 GDSII =0.9*(L130E GDSII) Dimension (unit : um)	L110 SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	(L sizing back 12 nm) 9/0.12
Salicide poly resistor	(Width/Length)	0.9*(Width/Length)	
3.3V I/O (W/L)	10/0.34	9/0.306	(L sizing back 12 nm) 9/0.318
contact size	0.16*0.16	0.144*0.144	
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	
Metal size	Width/Space	0.9*(Width/Space)	
Sloting inserting	metal line width>5.556um	mwtal line width>5um	
N-well,salicide and non-salicide resistor	(Width/Length)	0.9*(Width/Length)	

Note:

1. Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.
2. Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 3.3 V n-ch MOSFET

2.1 Silicon Wafer for modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	**MOSFET AC Model**	**Junction DC Model**	**Junction AC Model**
Mask ID	D1A0C9F	D1A0C9F	D1A0C9F	D1A0C9F
Lot ID	QGPLY	QGPLY	QGPLY	QGPLY
Wafer Number	#8	#8	#8	#8

2.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =300nA*W_{DES}*0.9/(L_{DES}*0.9+0.012µm). The dimension is after 90% shrink and 12nm sizing channel length back.

Table 2-2 Electrical Parameters for NMOS

All parameters are given for T=25 ^oC ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = 0.05 V	V	-0.135	-0.156	-0.123	-0.123
I_{ON}	V_{DS} = 3.3 V	μA/ μm	96	95.30	95.52	95.52
Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 1.8 μ m)						
V_{TLIN}	V_{DS} = 0.05 V	V	-0.16	-0.177	-0.147	-0.147
V_{TSAT}	V_{DS} = 3.3 V	V	-0.216	-0.235	-0.224	-0.224
Body Effect V_T shift	V_{DS} = 3.3 V V_{BS}= 0 ~ -3.3 V	V	--	0.066	0.068	0.068
DIBL V_T shift	V_{DS}= 0.05~3.3 V	V	0.056	0.058	0.077	0.077
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	402	404.54	409.54	409.54
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	--	4.3E-06	4.7E-06	4.7E-06
Narrow and long channel device (W_{DES}/L_{DES}= 1 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = 0.05 V	V	-0.108	-0.123	-0.097	-0.097
V_{TSAT}	V_{DS}= 3.3 V	V	--	-0.130	-0.115	-0.115
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	84.1	83.71	84.31	84.31
Narrow and short channel device (W_{DES}/L_{DES}= 1 μm / 1.8 μm)						
V_{TLIN}	V_{DS} = 0.05 V	V	-0.125	-0.140	-0.105	-0.105
V_{TSAT}	V_{DS}= 3.3 V	V	-0.141	-0.163	-0.154	-0.154
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	383	382.50	382.02	382.02
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	1.94e-6	1.8E-06	2.3E-06	2.3E-06
Capacitance						
CJS	V_J = 0 V	F/m^2	1.20e-4	1.20e-4	1.20e-4	1.20e-4
CJSWS	V_J = 0 V	F/m	3.45e-10	3.45e-10	3.45e-10	3.45e-10
CJSWGS	V_J = 0 V	F/m	1.12e-10	1.12e-10	1.12e-10	1.12e-10

2.3 MOSFET DC model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

W\L	10	5	3	2.5	2	1.8
10	X	X	X	X	X	X
5	X				X	X
2.5	X	X			X	X
1	X		X		X	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.1	3.3	3.2
V_{GS} (V)	-0.5	3.3	0.1
V_{BS} (V)	0	-3.3	-0.825

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	3.6	0.1
V_{GS} (V)	0.7	3.3	0.65
V_{BS} (V)	0	-3.3	-3.3

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (L_{DES}*0.9+12nm and W_{DES}*0.9) respectively.

- ***I_D vs. V_{DS} Plots***

The following plots show measured and simulated output I-V curves for selected devices at 25 ℃. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

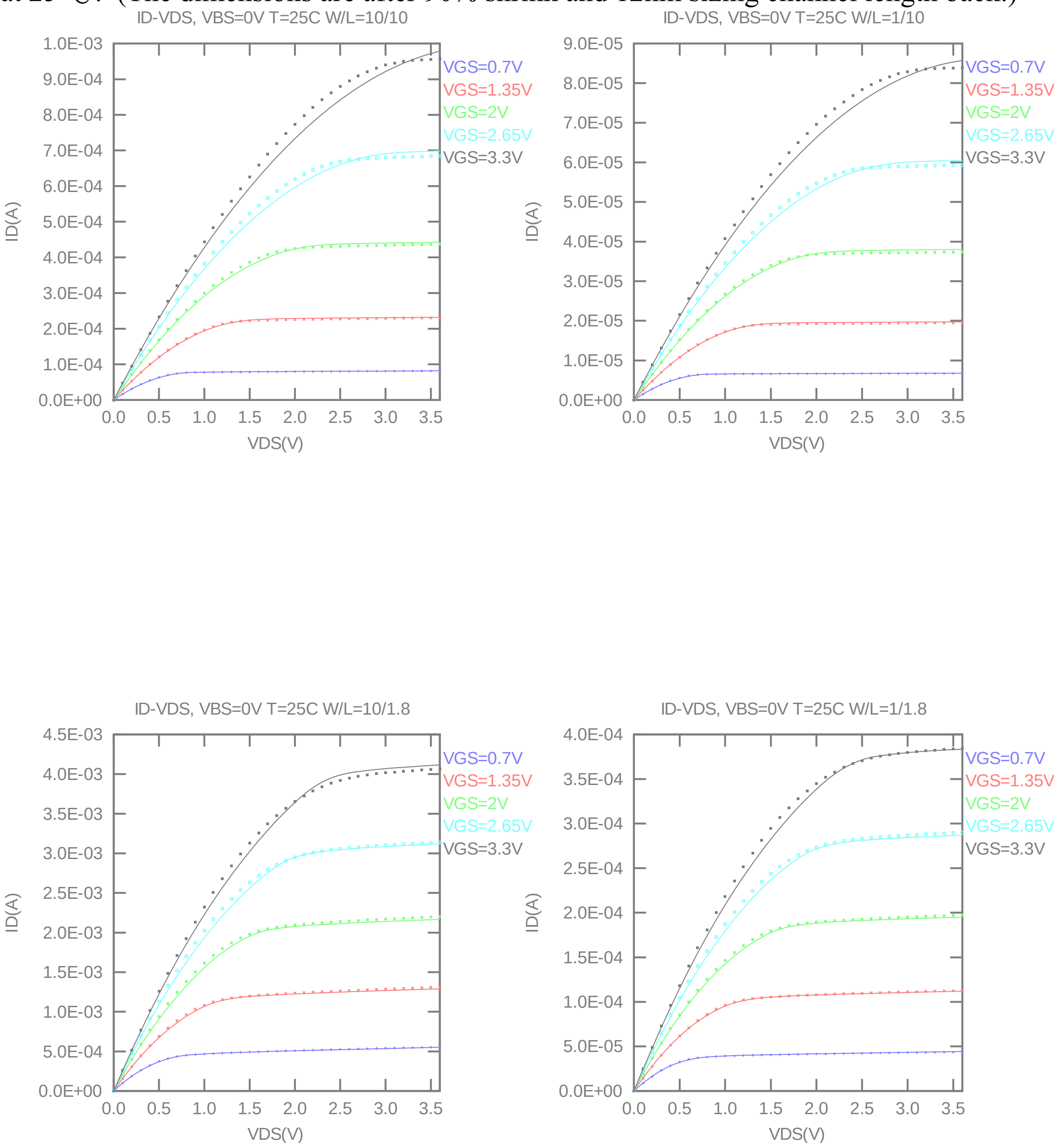

Figure 2-1 I_D vs. V_{DS} at V_{BS} = 0 V for NMOS

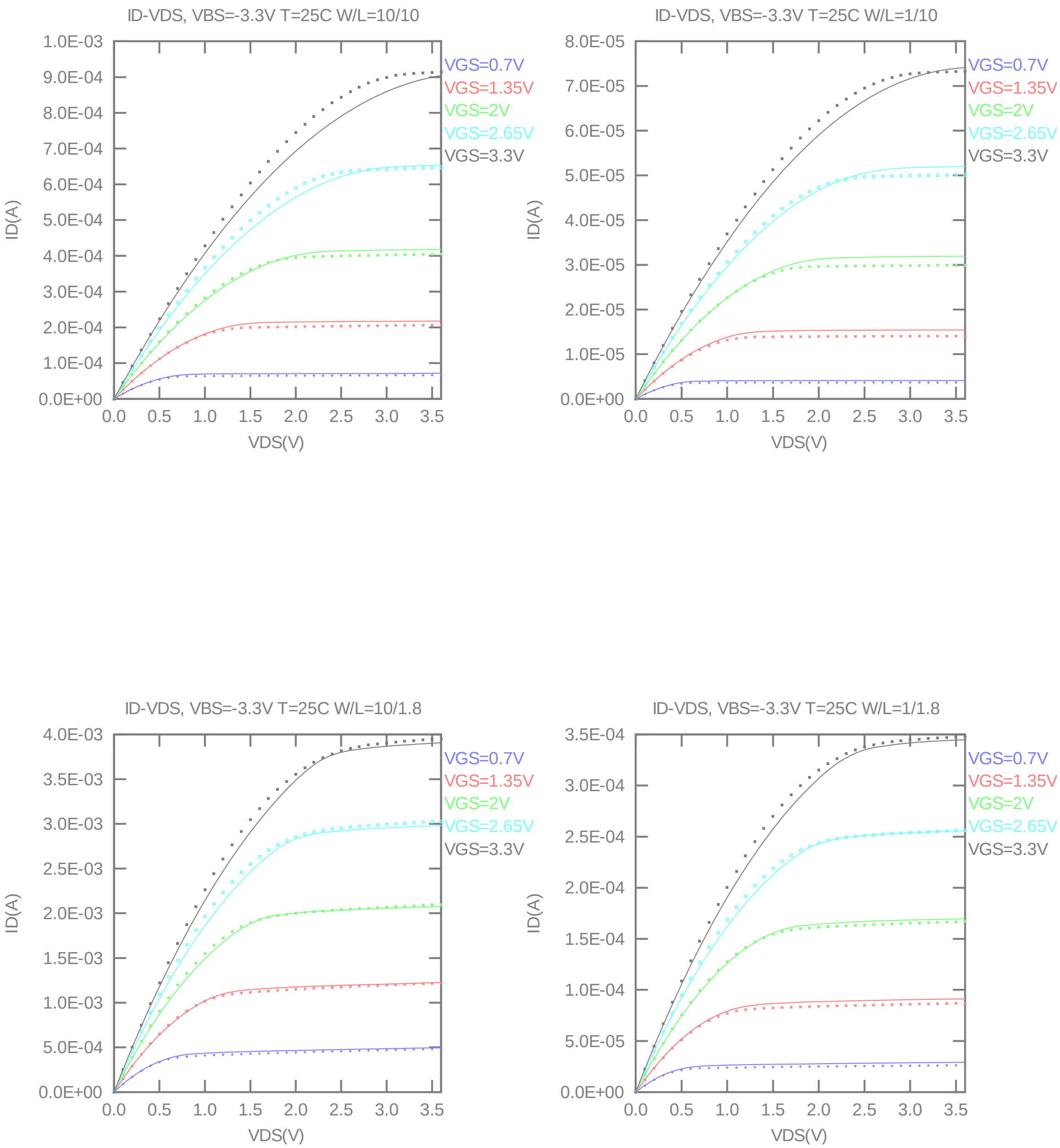

Figure 2-2 I_D vs. V_{DS} at V_{BS}=-3.3 V for NMOS

➢ ***G_{DS} vs. V_{DS} Plots***

The following plots show measured and simulated output conductance curves for selected devices at 25 ℃. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

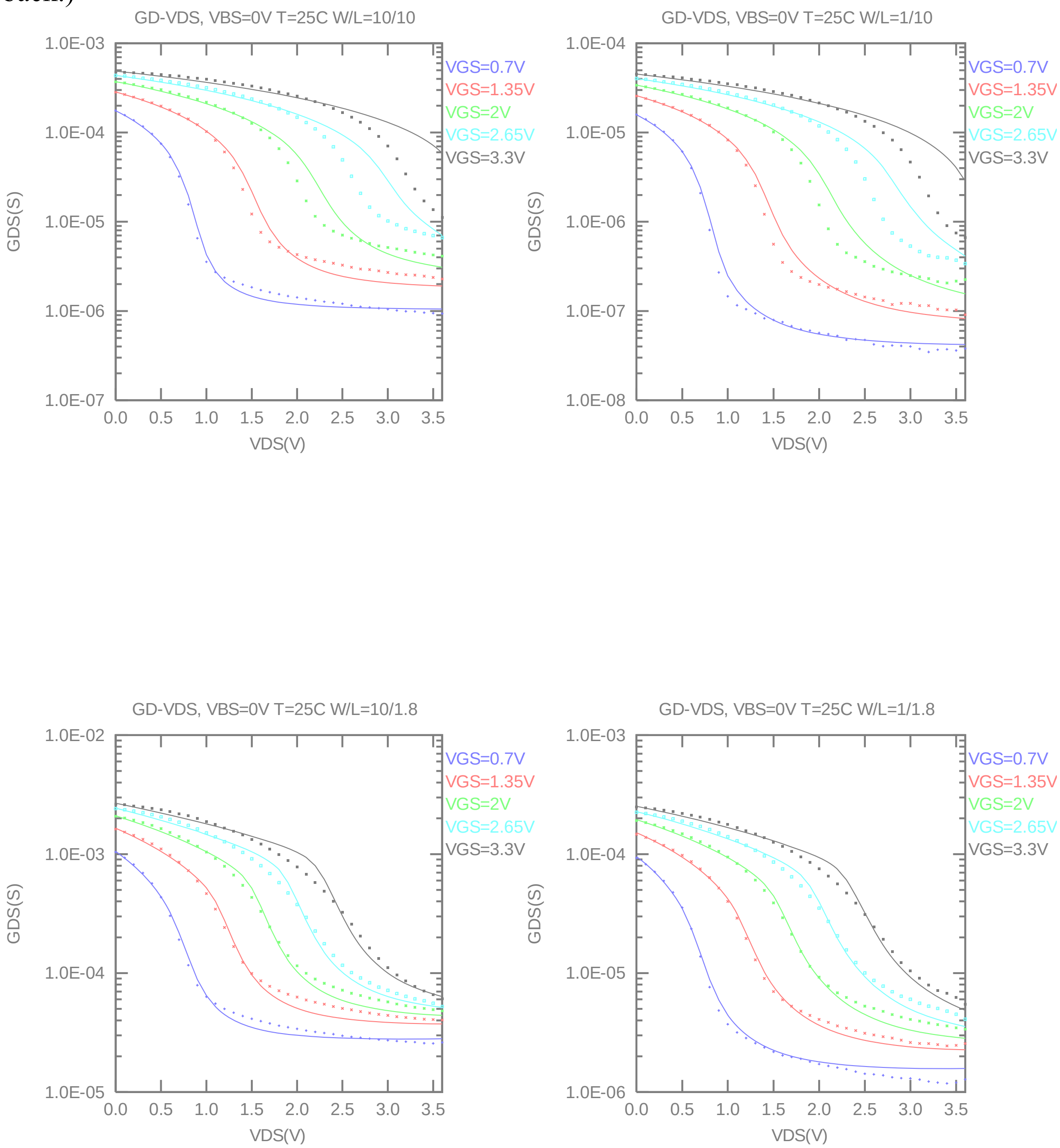

Figure 2-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for NMOS

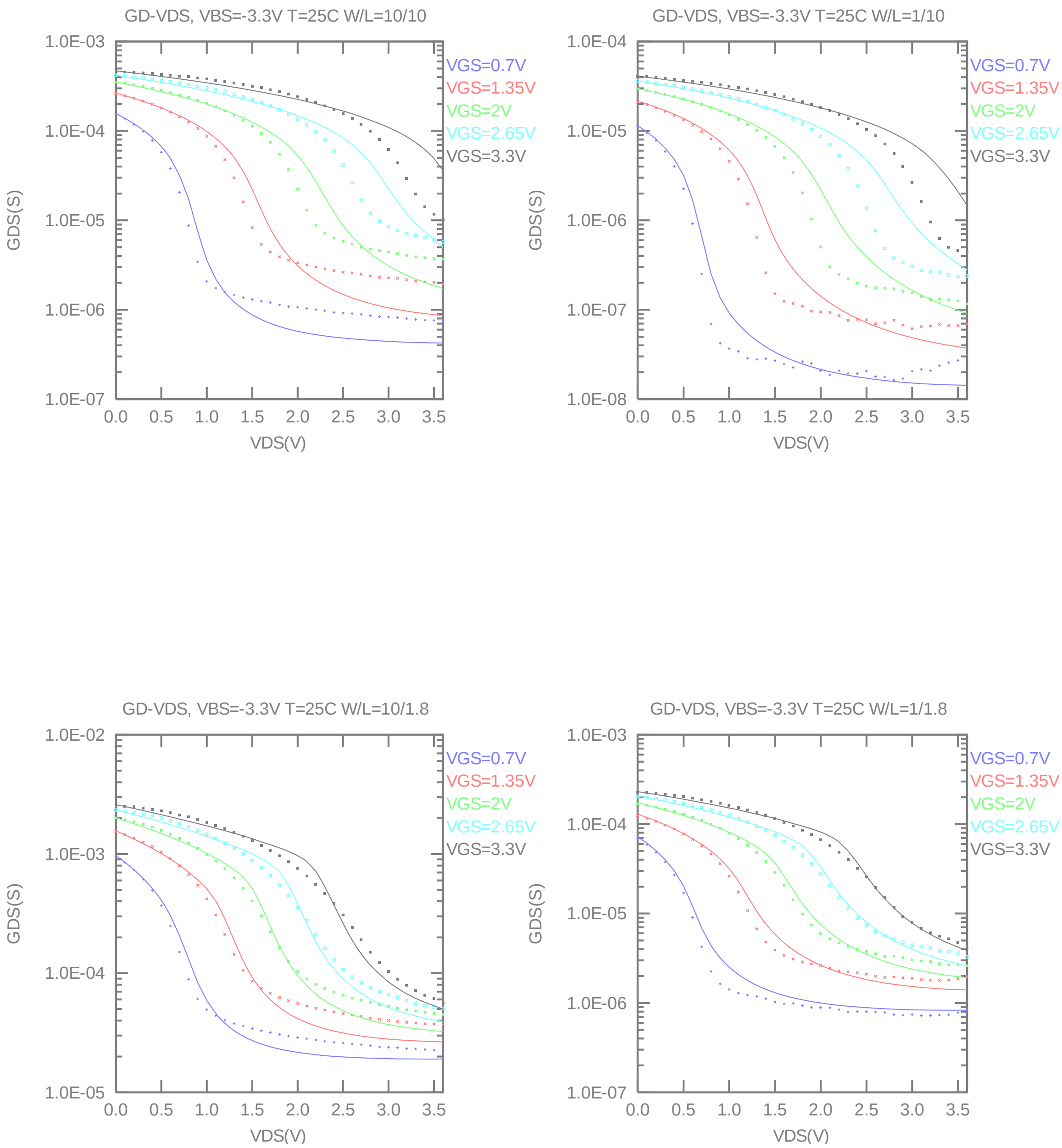

Figure 2-4 G_{DS} vs. V_{DS} at V_{BS} = -3.3 V for NMOS

➢ I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

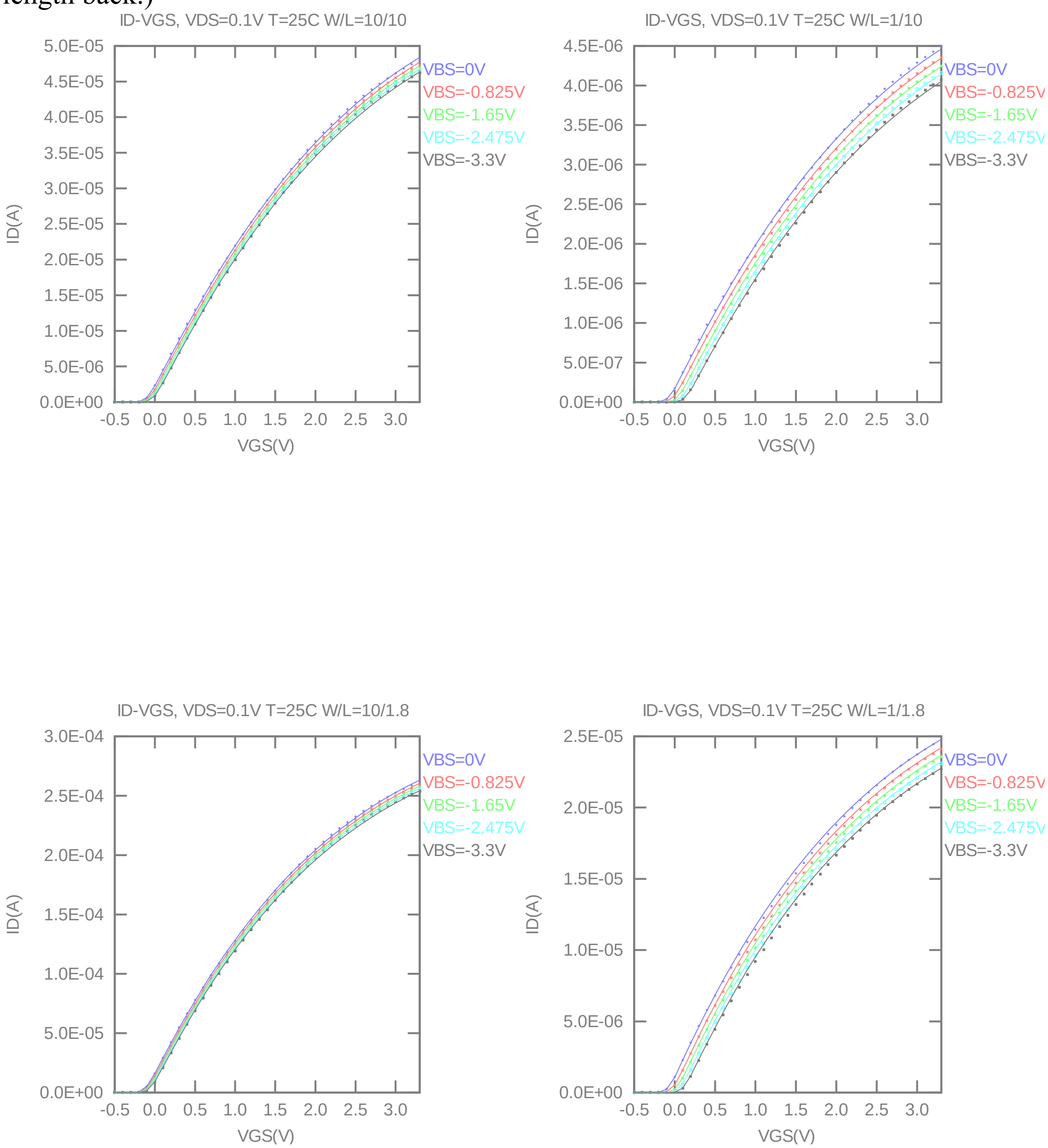

Figure 2-5 I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

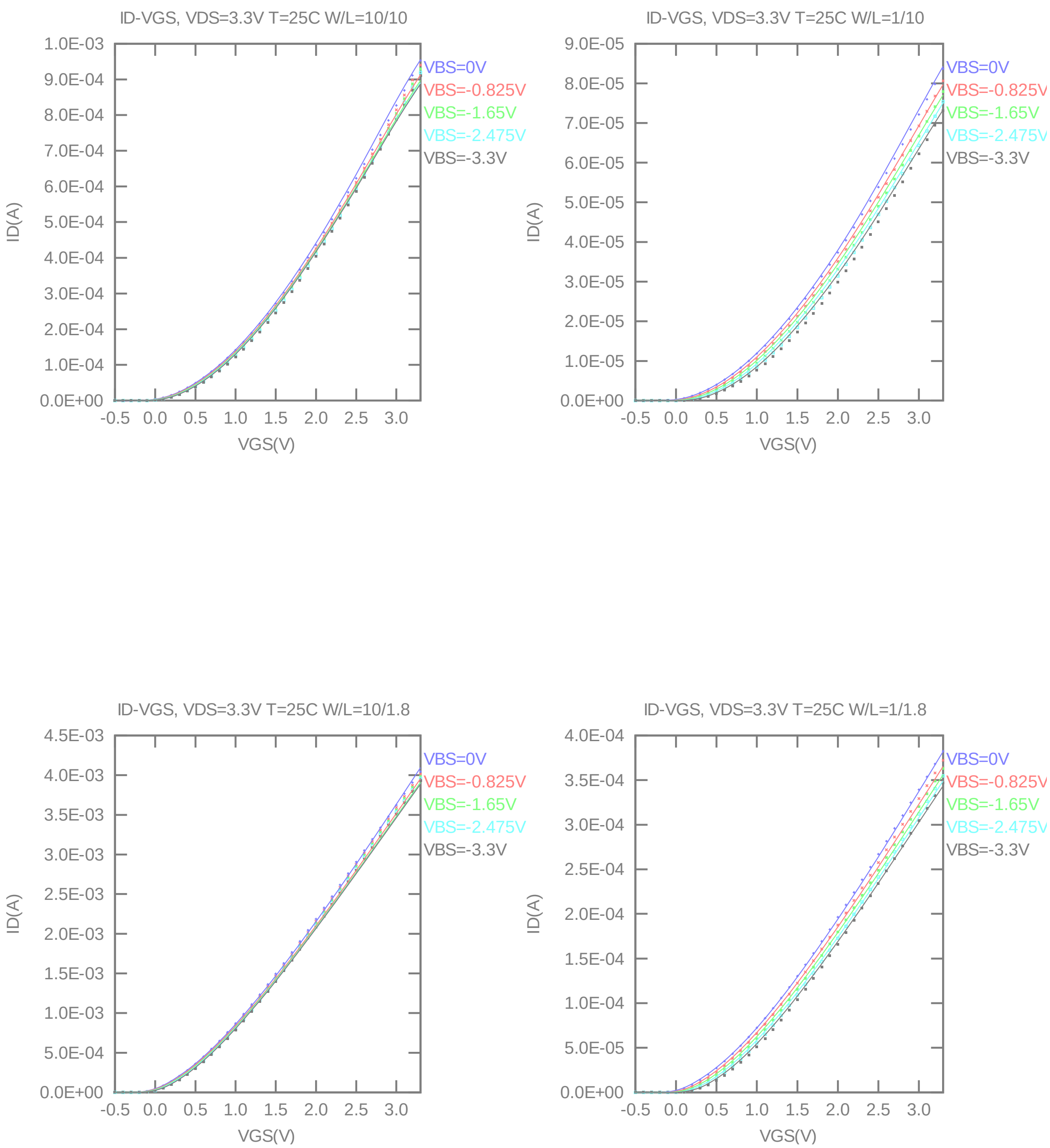

Figure 2-6 I_D vs. V_{GS} at V_{DS} = 3.3 V for NMOS

➢ I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

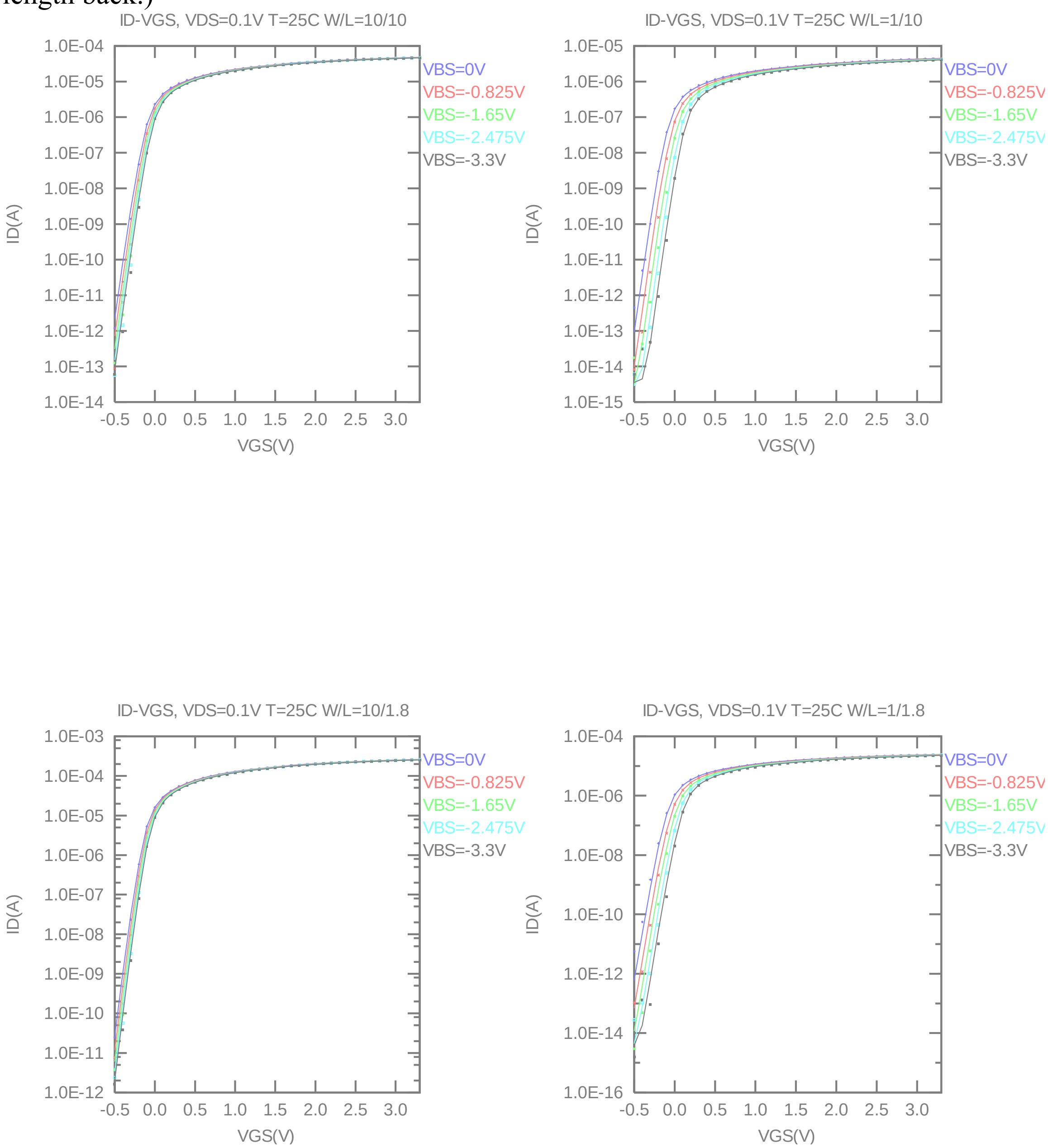

Figure 2-7 Log scaled I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

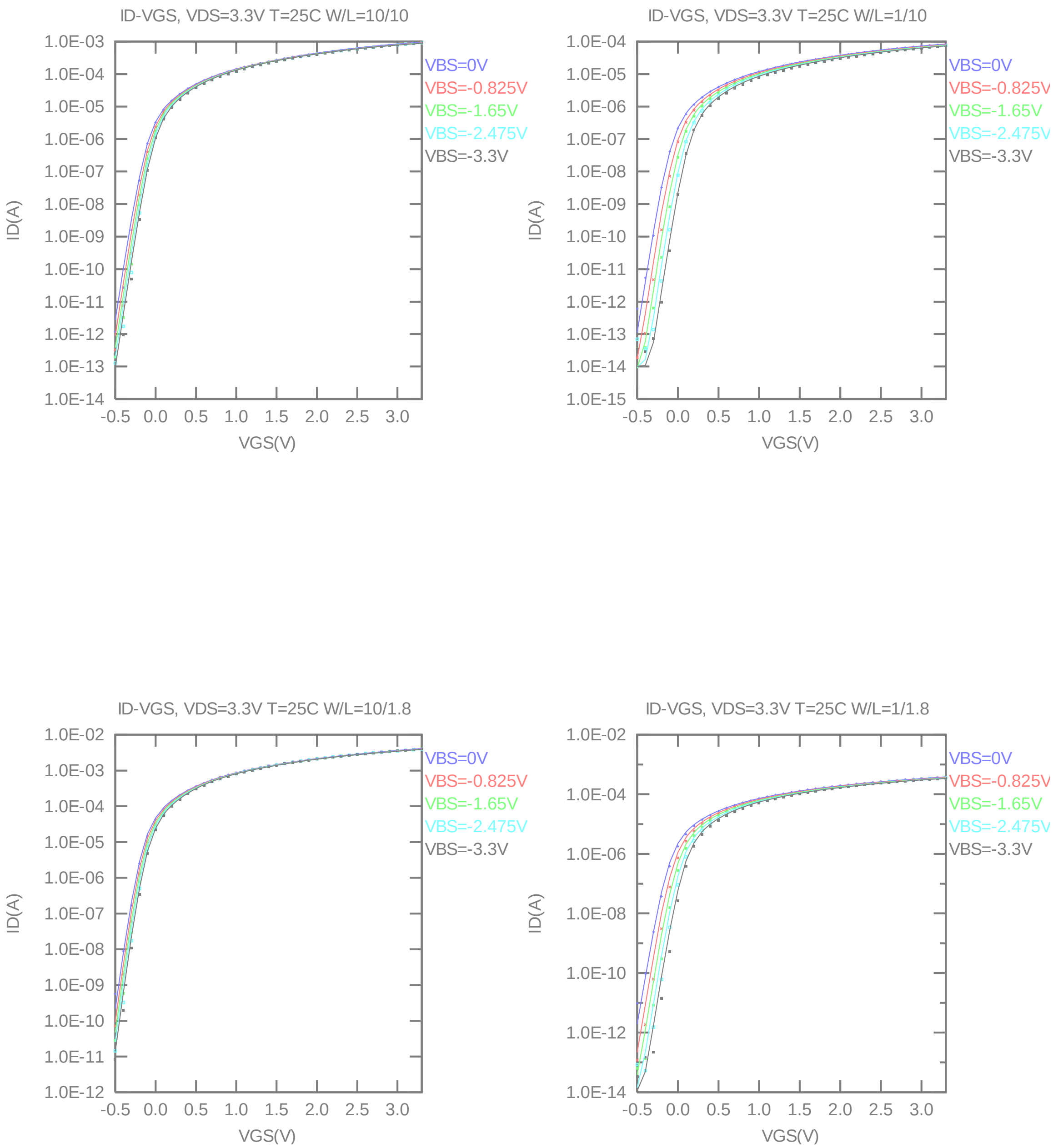

Figure 2-8 Log scaled I_D vs. V_{GS} at V_{DS} = 3.3V for NMOS

- ***G_M vs. V_{GS} Plots***

The following plots show measured and simulated transfer conductance curves in linear scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

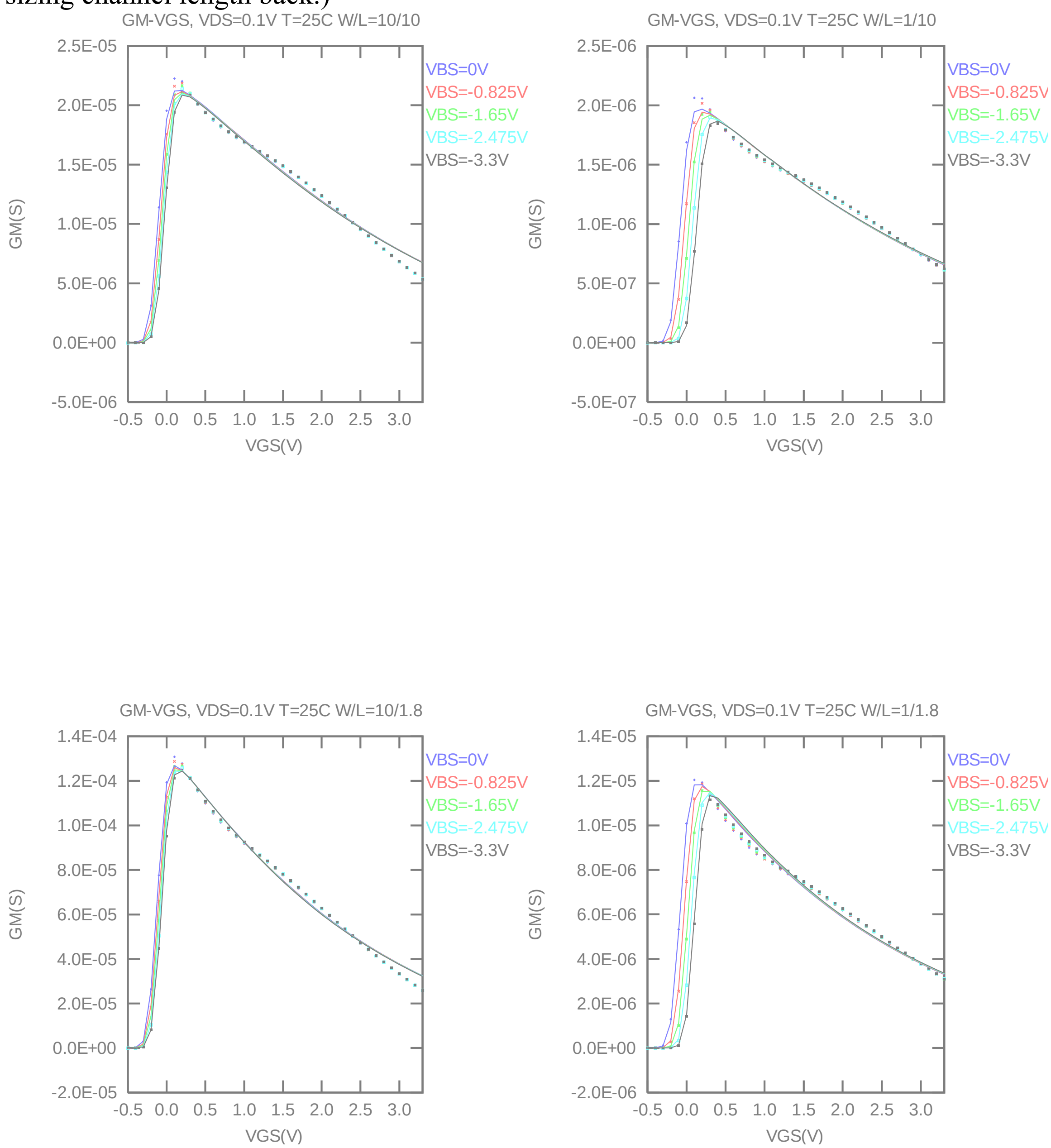

Figure 2-9 G_M vs. V_{GS} at V_{DS} = 0.1 V for NMOS

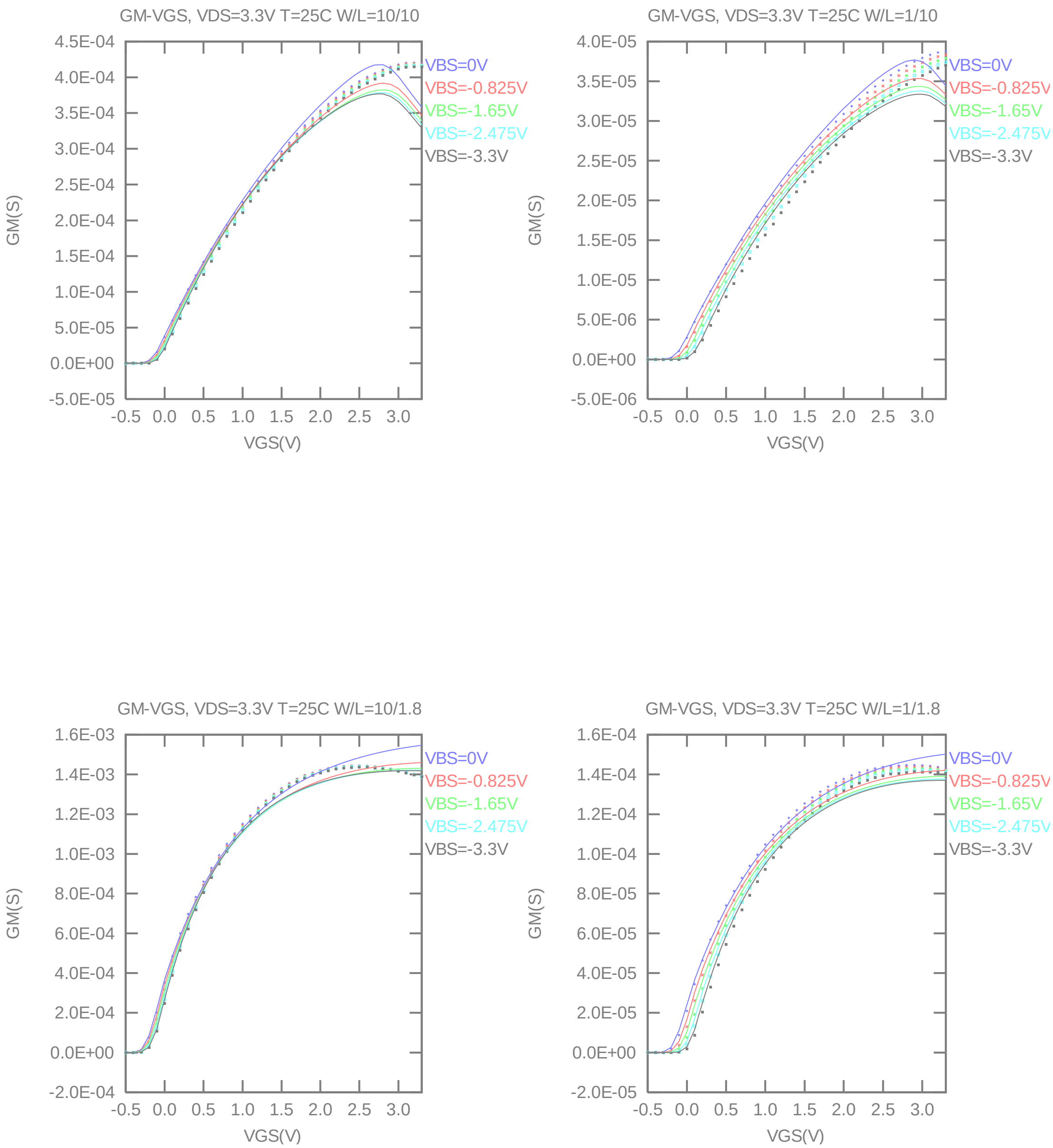

Figure 2-10 G_M vs. V_{GS} at V_{DS} = 3.3 V for NMOS

➢ ***V_T vs. L_{DES} & V_T vs. W_{DES}***

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =300nA*W_{DES}*0.9/(L_{DES}*0.9+0.012 µm) at 25 ℃. The plots show the physical dimension. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

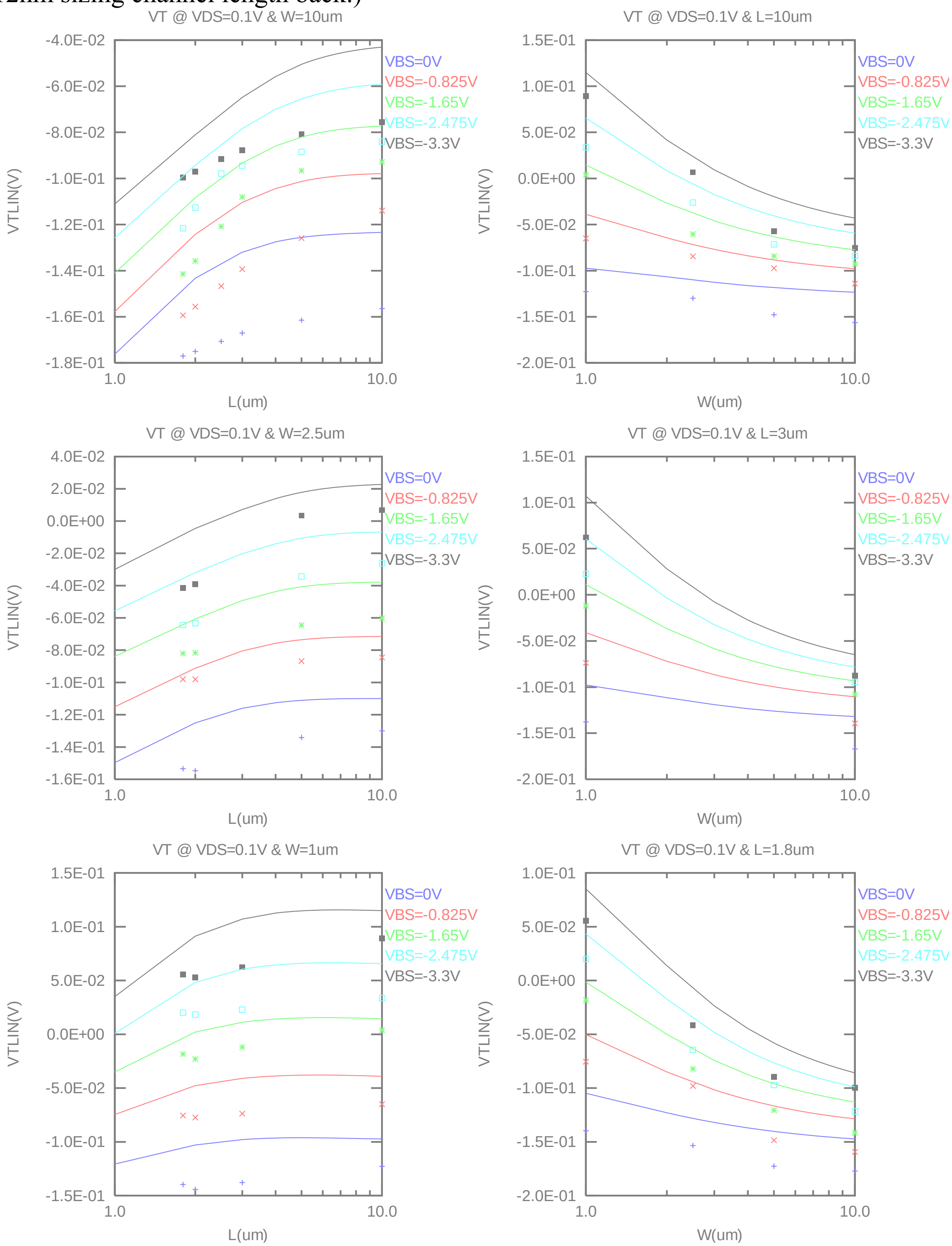

Figure 2-11 V_T vs. channel length & channel width for low drain bias (V_{DS} = 0.1 V) for NMOS

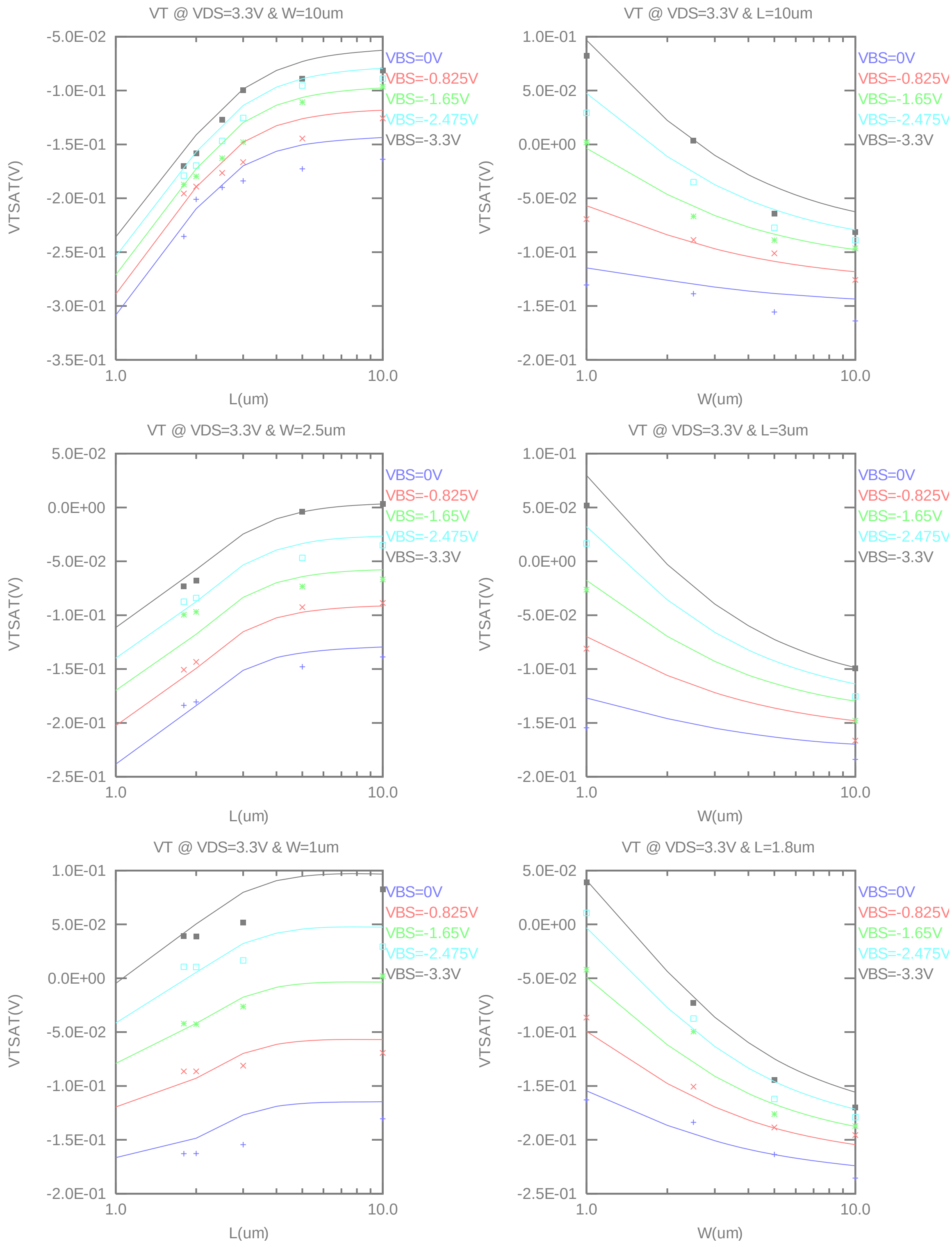

Figure 2-12 V_T vs. channel length & channel width for higt drain bias (V_{DS} = 3.3 V) for NMOS

➢ I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 1.2V$ for $V_{BS} = 0$ V and $V_{BS} = -1.2$ V at 25 ℃. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

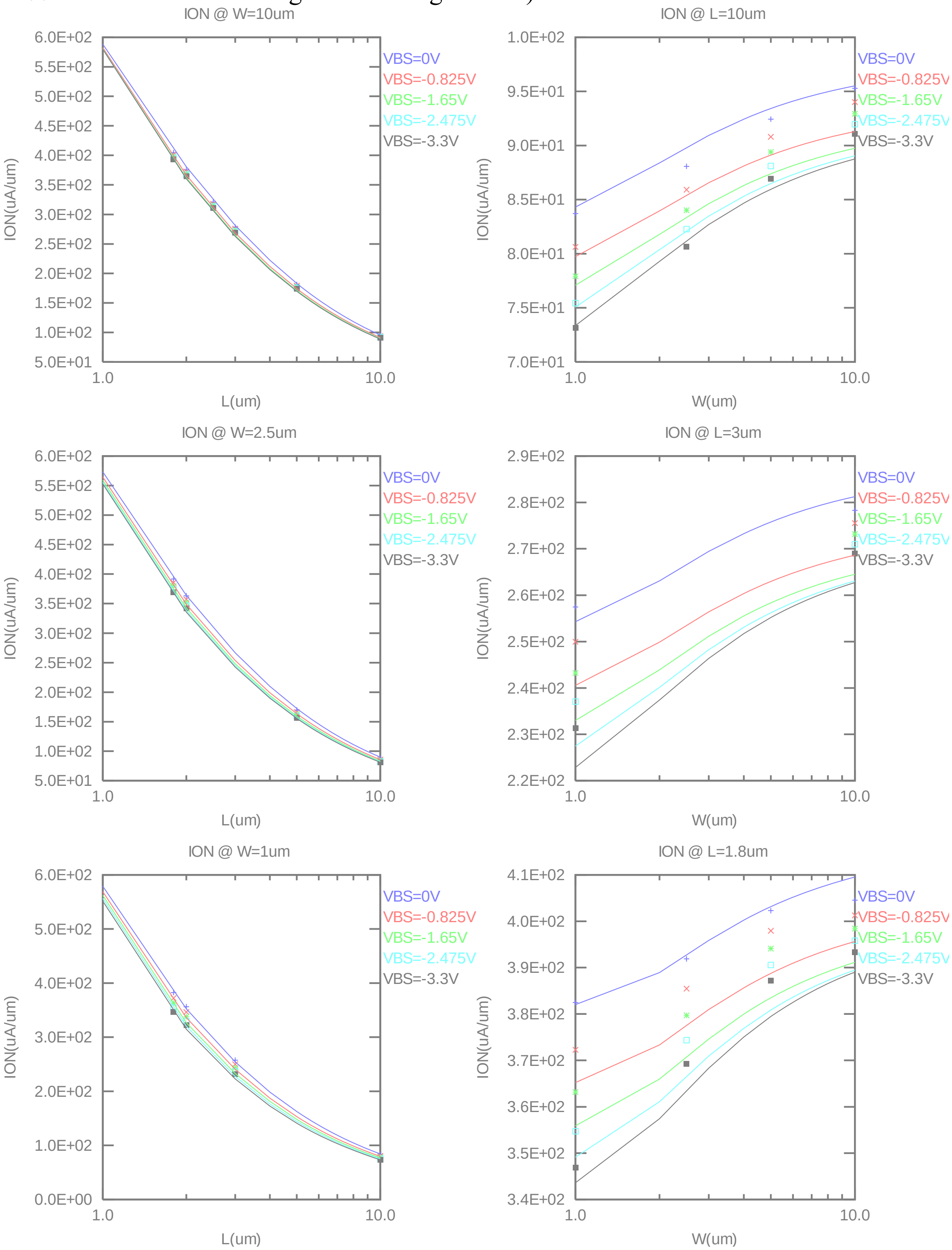

Figure 2-13 Saturation current I_{ON} vs channel length and channel width for NMOS

➢ V_T vs. Temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at I_D = 300nA*W_{DES}*0.9/(L_{DES}*0.9+0.012 µm). (The dimensions are after 90% shrink and 12nm sizing channel length back.)

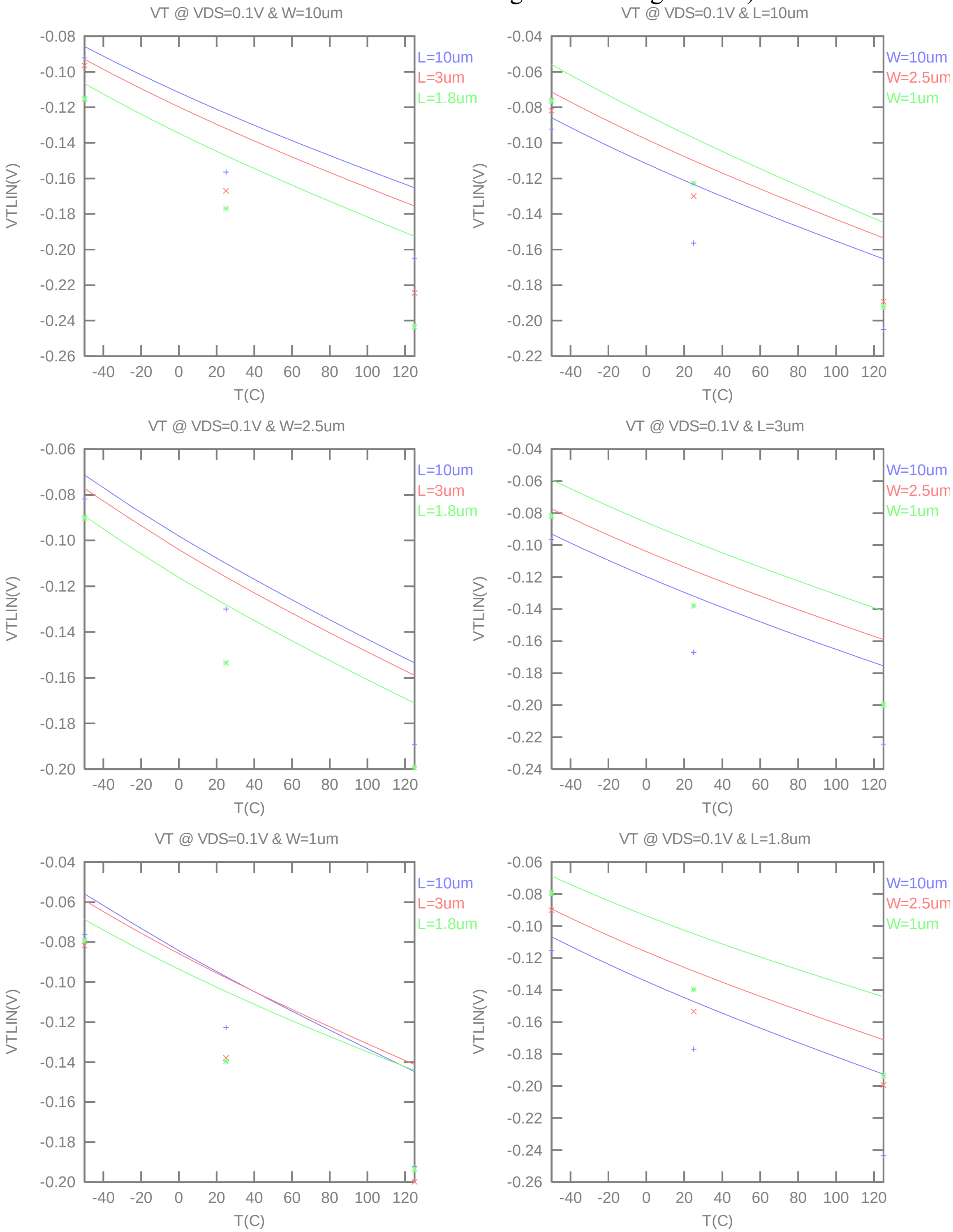

Figure 2-14 V_{TLIN} vs temperature for NMOS

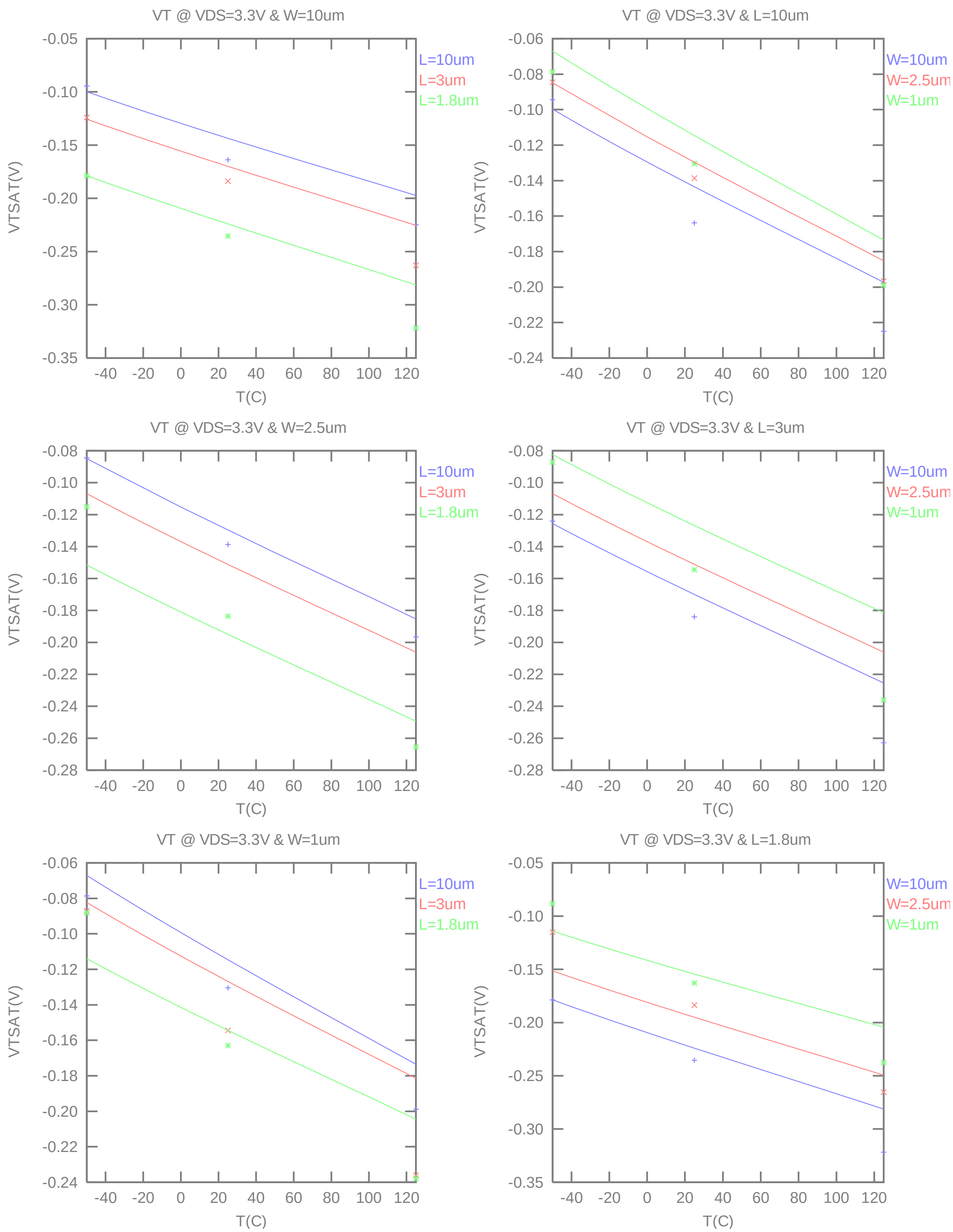

Figure 2-15 V_{TSAT} vs temperature for NMOS

➢ ***I_{ON} vs Temperature***

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 1.2$ V, $V_{BS} = 0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

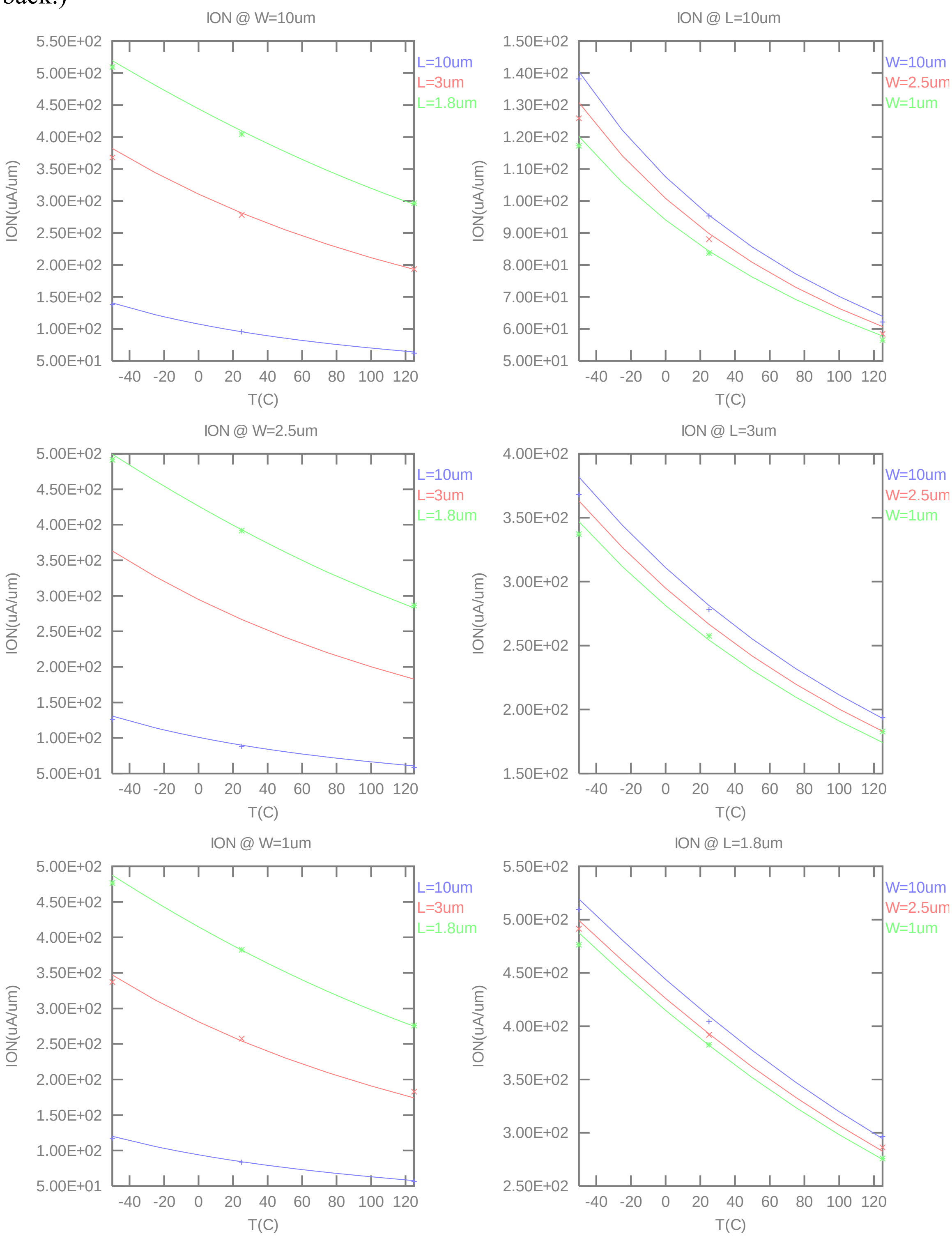

Figure 2-16 Saturation Current vs temperature for NMOS

2.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
b. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
c. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$, $V_{CC} = 3.3V$

2) G_M and G_{DS} accuracy criteria

a. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
b. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 2-5 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	≤75%	≤25%	≤7%
G_M	≤50%	≤20%	≤20%
G_{DS}	≤100%	≤50%	≤50%

➢ *I_D Accuracy*

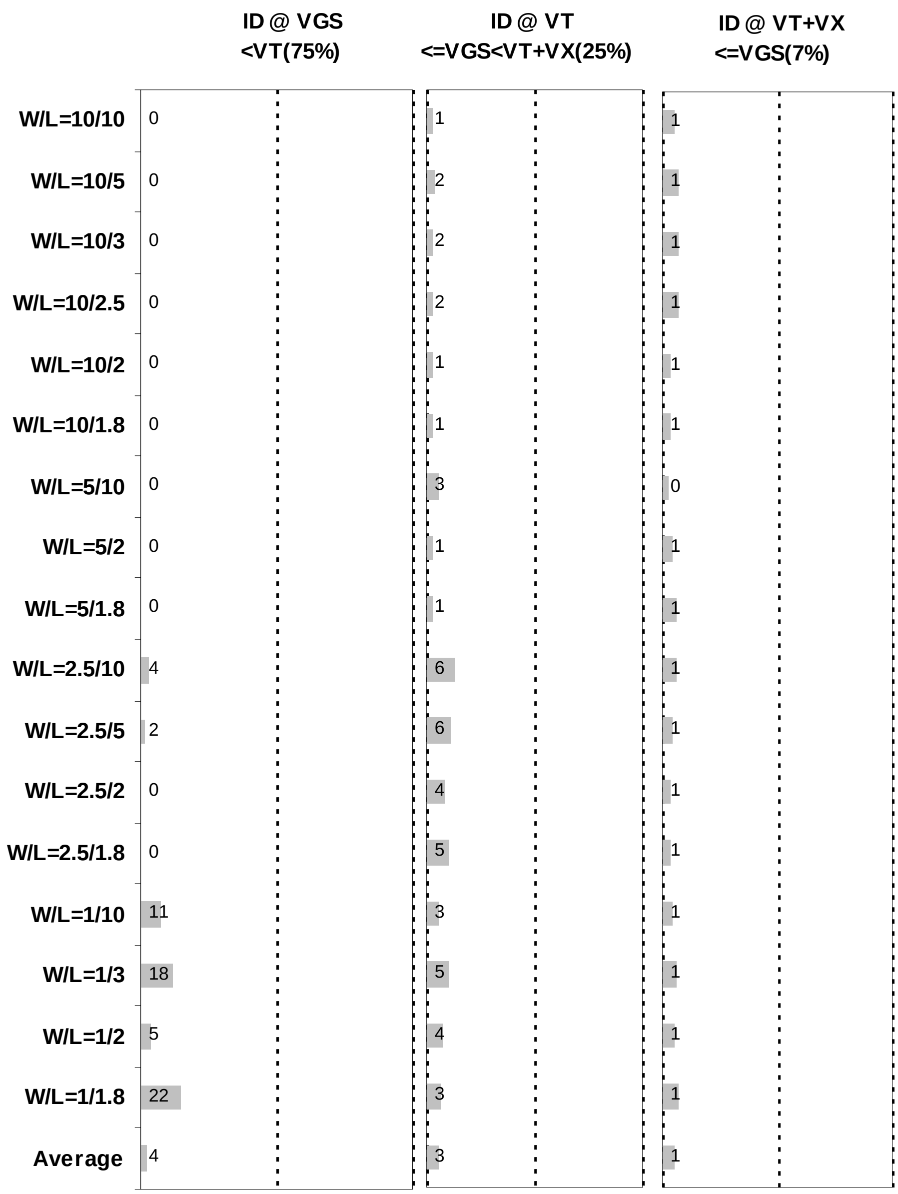

Figure 2-17 I_D accuracy at 25 ℃ for NMOS

➢ *G_M and G_{DS} Accuracy*

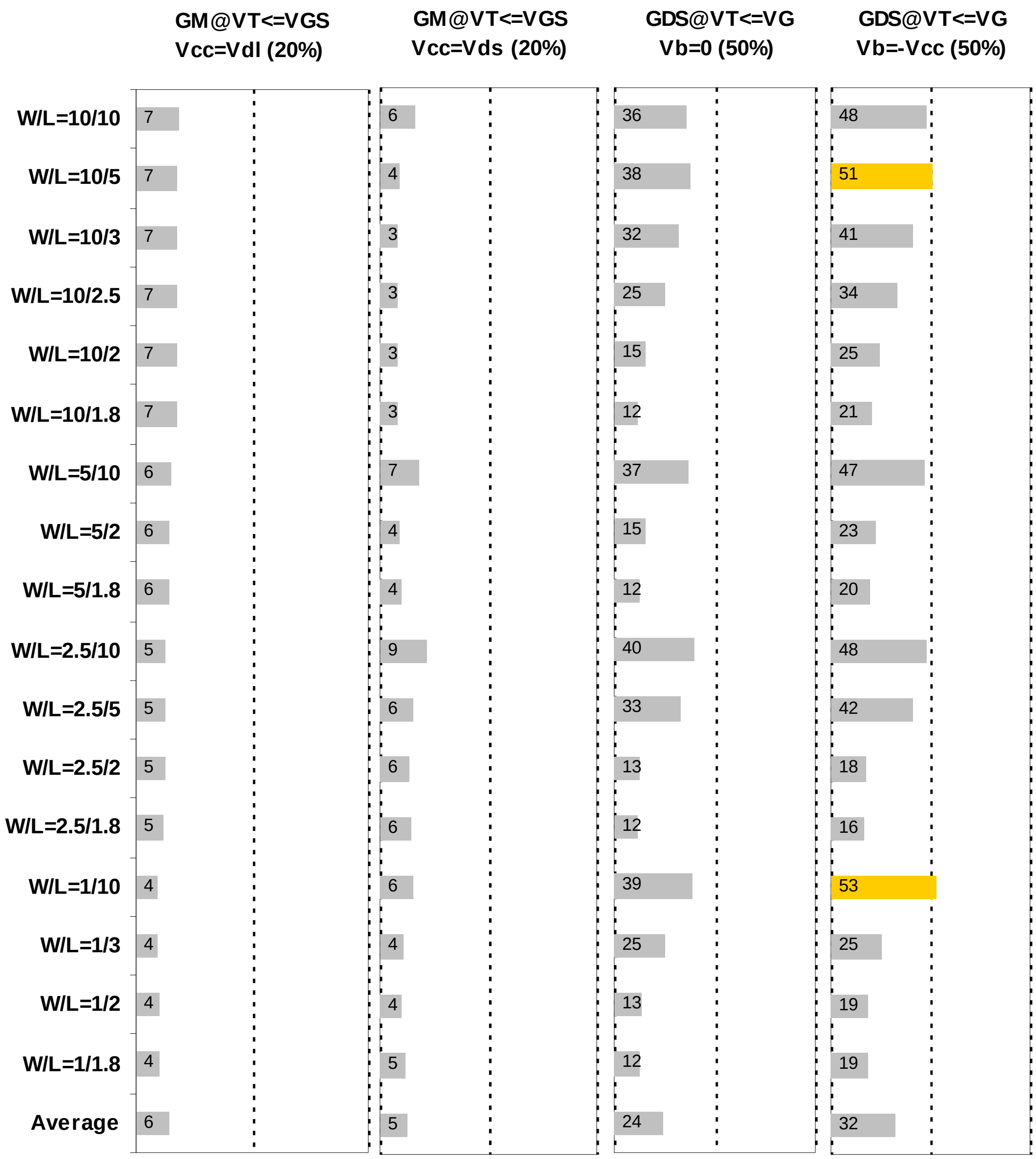

Figure 2-18 G_M and G_{DS} accuracy at 25 ℃ for NMOS

2.3.5 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structures for NMOS Cgg Capacitor structure

Structure	L_{DES}[µm]	W_{DES}[µm]
Gate capacitance components, C_{GDS}	100	200

2.3.6 Measurement conditions

Capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-7 capacitor measurement conditions for NMOS Cgg

V_{BIAS} [V]	From -3.3 to 3.3 V in 0.1 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.3.7 Tox extraction

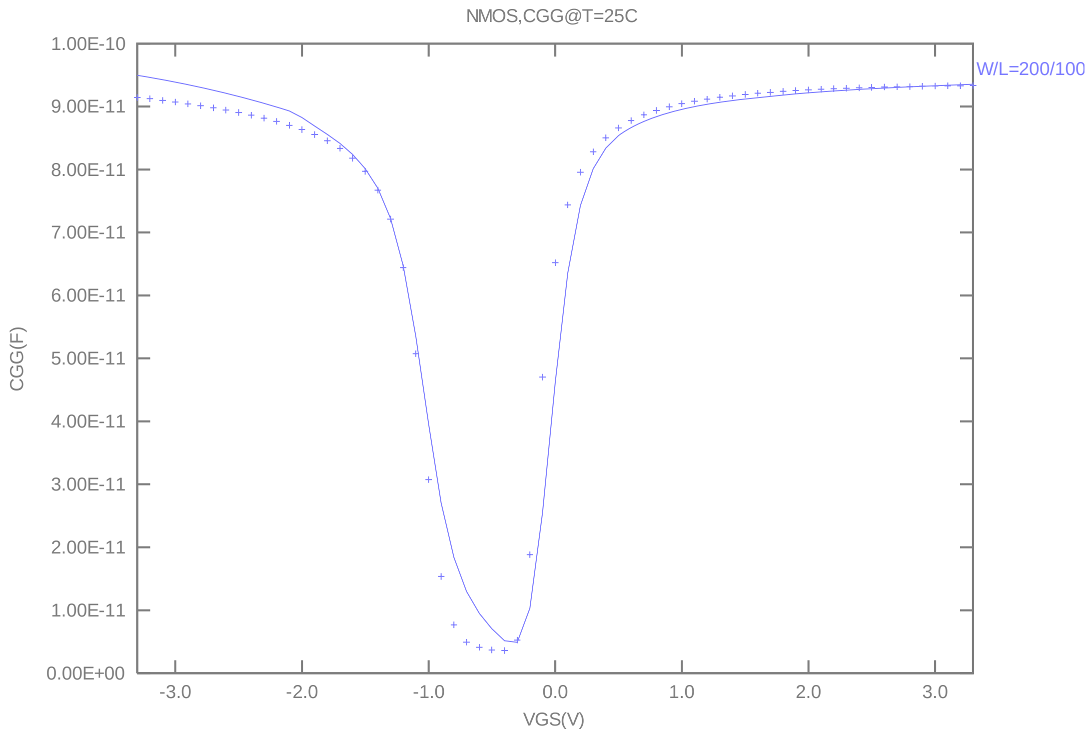

Figure 2-19 Measured and simulated Cgg curve

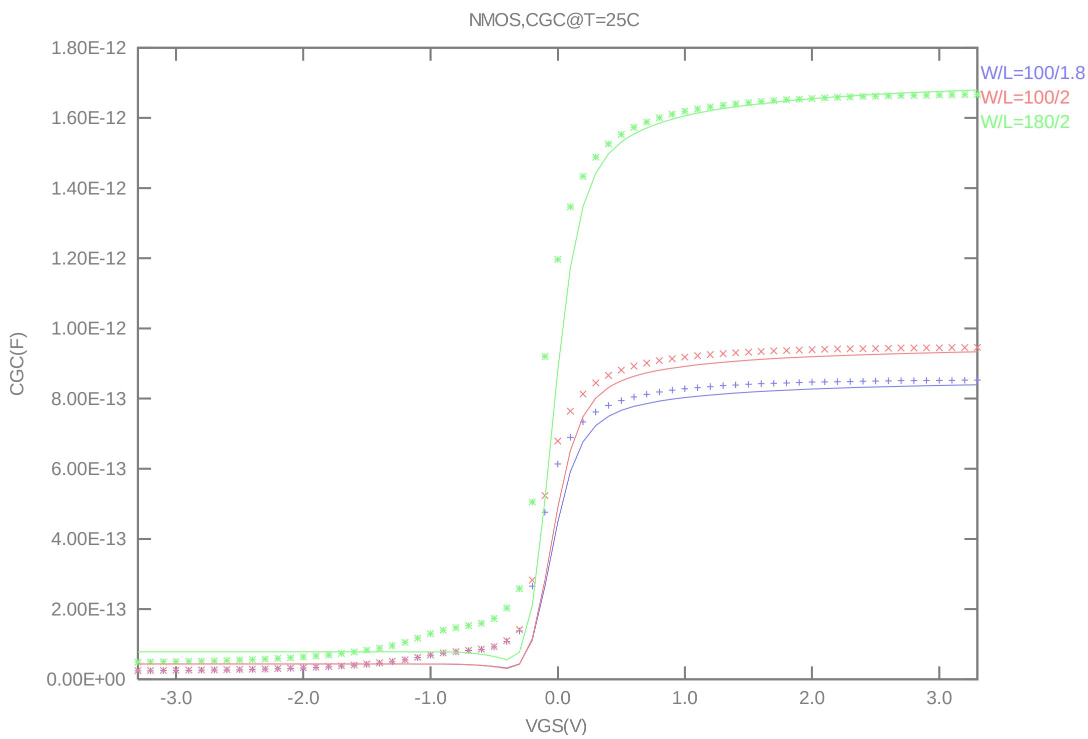

Figure 2-20 Measured and simulated Cgc curve

2.4 Parasitic source / drain junction capacitance model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-8 Junction capacitor structures for NMOS

Structures	Area[m^2]	Perimeter[m]	Description
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	2.70E-08	1.83E-02	Finger diode
Inner gate sidewall junction	0.00E+00	0.00E+00	Poly finger structure

2.4.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-9 Junction capacitor measurement conditions for NMOS

V_{BIAS} [V]	VH: pwell & VL: n+ From 0 to 3.6 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

2.4.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 2-10 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	**Unit**	**Value**	**Parameter**	**Unit**	**Value**	**Parameter**	**Unit**	**Value**
CJ	F/m^2	1.21E-04	CJSW	F/m	3.45E-10	CJSWG	F/m	1.13E-10
PB	V	4.66E-01	PBSW	V	8.40E-01	PBSWG	V	4.63E-01
MJ	--	3.15E-01	MJSW	--	1.00E-01	MJSWG	--	3.75E-01
TCJ	1/K	2.26E-03	TCJSW	1/K	5.37E-04	TCJSWG	1/K	2.35E-03
TPB	V/K	2.60E-03	TPBSW	V/K	4.30E-03	TPBSWG	V/K	2.60E-03

➢ *CJ, CJSW vs. VBIAS Plots*

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

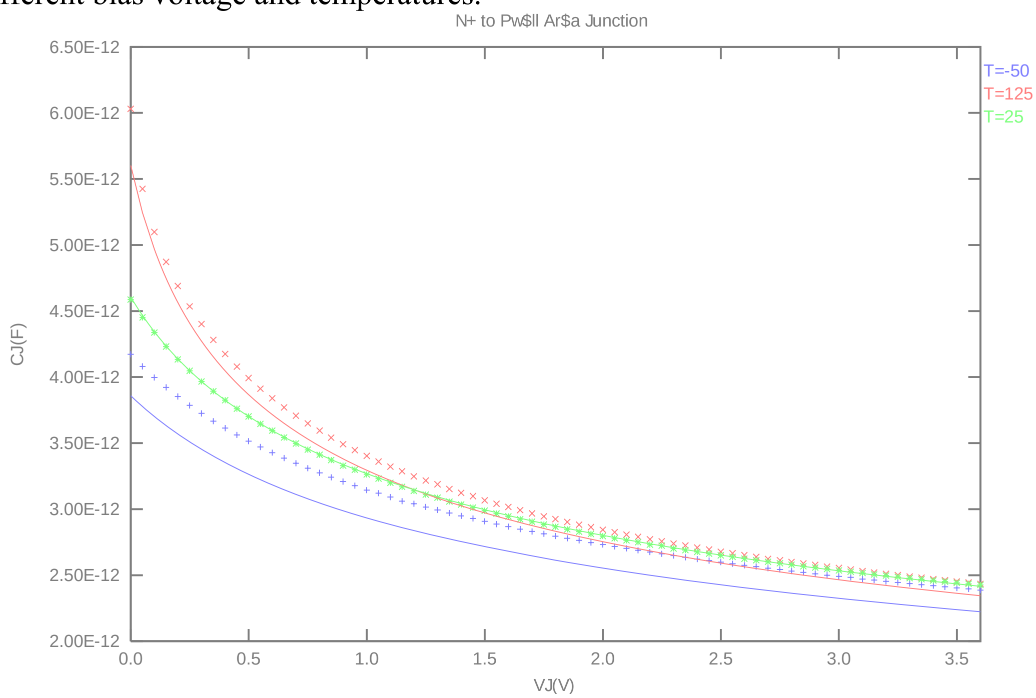

Figure 2-21 Bottom area junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for NMOS

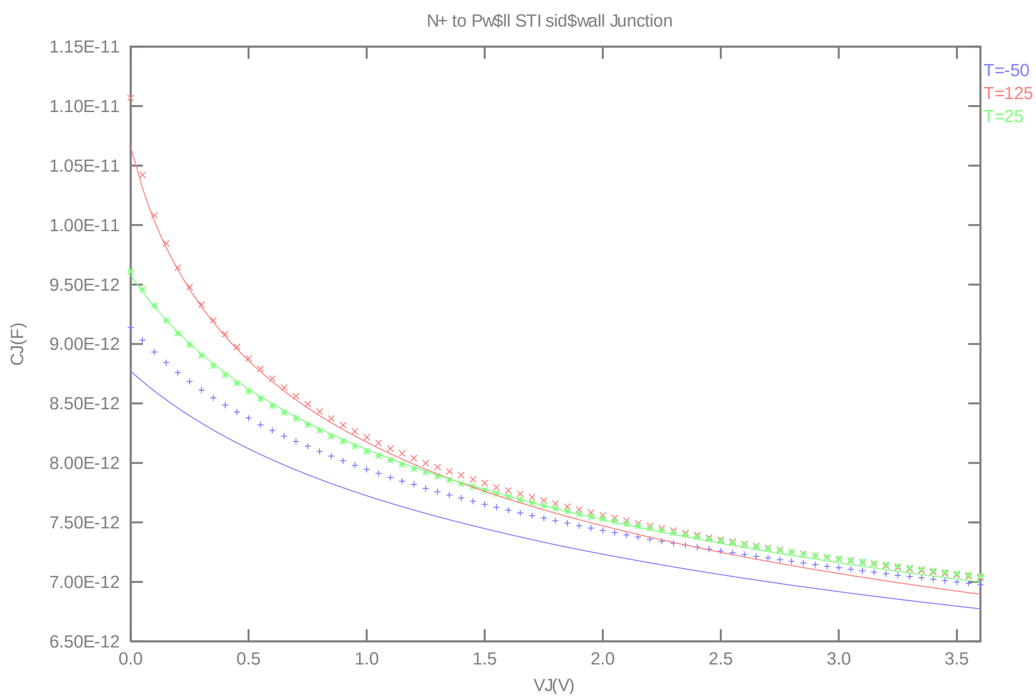

Figure 2-22 STI bounded sidewall junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for NMOS

3 Worst case model

3.1 Parameter Variations

Table 3-1 Corner parameters

	n_nvt_33_l110e	TT	SS	FF
1	tox	7.15E-09	+1.50E-10	-1.50E-10
2	wwl	-3.42E-22	-4.43E-23	+3.02E-23
3	xl	-1.30E-08	+1.00E-08	-1.00E-08
4	xw	-1.80E-08	-1.12E-08	+6.77E-09
5	vth0	-1.44E-01	+6.82E-02	-6.82E-02
6	lvth0	0.00E+00	+3.70E-08	-4.15E-08
7	pvth0	3.66E-14	-3.98E-14	+4.07E-14
8	k3	1.80E+01	+4.82E+00	-5.21E+00
9	vsat	9.90E+04	+9.94E+03	-3.07E+03
10	rdsw	5.25E+02	+4.20E+02	-3.38E+02
11	u0	4.69E-02	-1.49E-03	+1.54E-03
12	lu0	9.66E-11	-4.10E-09	+3.73E-11
13	wu0	-3.00E-09	-3.49E-10	+2.74E-10
14	ags	1.27E+00	+7.67E-03	-2.97E-02
15	wags	0.00E+00	+1.31E-07	-1.28E-07
16	pu0	0.00E+00	+1.27E-15	-4.22E-16
17	pags	0.00E+00	-6.10E-13	+6.39E-13
18	cj	1.21E-04	+10.0%	-10.0%

19	cjsw	3.45E-10	+10.0%	-10.0%
20	cjswg	1.13E-10	+10.0%	-10.0%
21	cgdo	2.16E-10	-10.0%	+10.0%
22	cgso	2.16E-10	-10.0%	+10.0%

- ***DEVICE PERFORMANCE LOOK UP TABLE (SIMULATION RESULTS FROM HSPICE VER.05.3):***

** The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 12nm. It means: The EDR value of size W/L : 10um/1.8um would be the same as the model card simulation result in size of W/L : 9um/1.62um(After 90% shrink without 12nm sizing channel length back).

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
VTH (V)	VDS	0.05V	3.3V	0.05V	3.3V	0.05V	3.3V
TT	9/9	-0.085	-0.100	-0.123	-0.144	-0.165	-0.197
SS	9/9	-0.006	-0.021	-0.042	-0.063	-0.081	-0.115
FF	9/9	-0.165	-0.179	-0.204	-0.224	-0.248	-0.279
TT	9/1.62	-0.109	-0.191	-0.150	-0.237	-0.196	-0.294
SS	9/1.62	-0.009	-0.091	-0.047	-0.135	-0.089	-0.189
FF	9/1.62	-0.210	-0.292	-0.253	-0.338	-0.302	-0.398
TT	0.9/9	-0.054	-0.065	-0.096	-0.113	-0.144	-0.172
SS	0.9/9	0.045	0.034	0.005	-0.013	-0.040	-0.070
FF	0.9/9	-0.154	-0.165	-0.198	-0.214	-0.249	-0.275
TT	0.9/1.62	-0.069	-0.114	-0.104	-0.153	-0.142	-0.200
SS	0.9/1.62	0.028	-0.018	-0.005	-0.055	-0.039	-0.099
FF	0.9/1.62	-0.167	-0.211	-0.203	-0.251	-0.244	-0.301

Table 3-2 Corner window of NMOS Vtln

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (μA/um)	VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	51.09	154.42	33.96	105.27	22.33	70.48
SS	9/9	43.67	138.09	29.19	94.55	19.34	63.71
FF	9/9	58.97	170.90	38.97	115.97	25.47	77.23
TT	9/1.62	227.36	544.91	171.48	436.74	118.66	318.26
SS	9/1.62	188.98	482.88	140.11	383.11	96.01	276.36
FF	9/1.62	267.76	612.90	202.56	490.50	140.08	356.32
TT	0.9/9	42.56	132.10	29.53	93.01	20.22	63.69
SS	0.9/9	34.80	115.09	24.32	81.52	16.87	56.55
FF	0.9/9	50.91	149.26	35.08	104.50	23.81	70.87
TT	0.9/1.62	204.98	517.03	154.60	412.64	108.98	300.88
SS	0.9/1.62	165.61	443.00	123.20	348.83	86.30	251.16
FF	0.9/1.62	246.30	595.22	186.44	476.33	131.30	347.84

Table 3-3 Corner window of NMOS Ion

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IOFF (A)	VDS	0.05V	3.3V	0.05V	3.3V	0.05V	3.3V
TT	9/9	1.29E-06	2.58E-06	1.35E-06	3.39E-06	1.34E-06	4.32E-06
SS	9/9	3.42E-07	5.11E-07	5.74E-07	1.06E-06	7.25E-07	1.80E-06
FF	9/9	2.69E-06	7.26E-06	2.35E-06	7.62E-06	2.08E-06	8.31E-06
TT	9/1.62	9.97E-06	5.42E-05	9.67E-06	5.36E-05	9.01E-06	5.48E-05
SS	9/1.62	2.03E-06	1.38E-05	3.43E-06	1.86E-05	4.25E-06	2.33E-05
FF	9/1.62	2.18E-05	1.21E-04	1.77E-05	1.07E-04	1.47E-05	9.94E-05
TT	0.9/9	8.24E-08	1.31E-07	1.04E-07	2.17E-07	1.18E-07	3.26E-07
SS	0.9/9	9.31E-09	1.16E-08	2.73E-08	3.92E-08	4.82E-08	9.45E-08
FF	0.9/9	2.34E-07	5.55E-07	2.19E-07	6.43E-07	2.08E-07	7.63E-07
TT	0.9/1.62	5.96E-07	1.92E-06	6.58E-07	2.28E-06	6.71E-07	2.67E-06
SS	0.9/1.62	8.31E-08	2.64E-07	1.81E-07	5.19E-07	2.67E-07	8.41E-07
FF	0.9/1.62	1.57E-06	5.97E-06	1.36E-06	5.75E-06	1.20E-06	5.80E-06

Table 3-4 Corner window of NMOS Ioff

4 Appendix

4.1 HSPICE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

4.2 SPECTRE Warning Message

Item	Warning Message	Explanation
1.	None.	Ignore

4.3 ELDO Warning Message

Item	Warning Message	Explanation
1.	None.	None.